



System Development Plan
for the
The Balance Project
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DOCUMENT CHANGE HISTORY

The following table is a simple list of released revisions sent for review. Records of reviews and the review artifacts are saved with reviewer information in the The Balance Project artifact repository.

Change Record

Date	Version	Author(s)	Change Reference
13 Sep 2025	P1	Vinay Agarwal	Preliminary DRAFT version
5 Dec 2025	P1	Vinay Agarwal	Final Draft of the system

Draft P1 Preliminary version of this document.

1. Baseline Document draft 1
2. Final fix and update from the first draft



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CHAPTER 1

Scope

This document provides the System Development Plan (**SDP**) for the Balance System. The system will be referred to as the Balance System.

1.1 Identification

The Balance System described in this document shall be known as Balance System version 1.

1.2 System Overview

Figure 1 shows the high-level architecture for the Balance System system. This diagram shows the major external interfaces that provide the capabilities of Balance System.

The Balance System is a single player game. The game is a standalone game played on the STM32 board. This system would be a game where the user would have to balance a ball on a LCD screen that is builtin on the STM32 board. The objective of the game is to balance the ball on the screen based on the way the board was tilted. Using an onboard gyroscope that communicates across the board to the MCU though SPI communication. Balance System would keep track of the current position of the ball and where the next updated move is. This helps keep track of the system of where the ball is until a movement has occurred. Balance System shall process at a maximum 180 Hz. This would give the user enough time to process the current angle of the ball and be able to present on the LCD-TFT screen.

1.3 Document Overview

This document contains an overview of how the software for the Balance System will be created and distributed to the hardware. It include system purpose, software development plan, and development procedure.

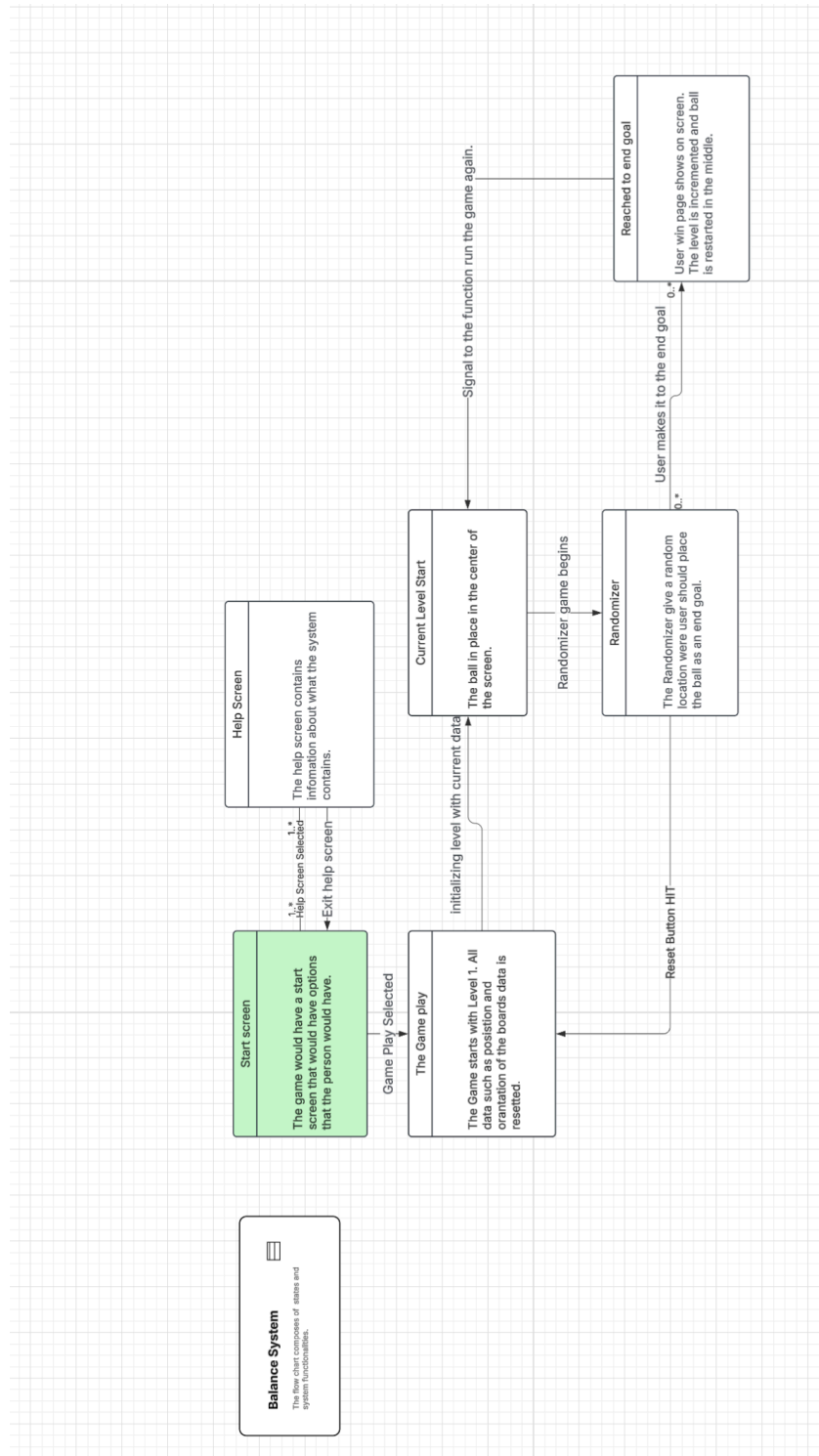


Figure 1: System Overview



CHAPTER 2

References

This section provides a list of referenced items for this document.

2.1 Acronyms and Abbreviations

This section defines acronyms and abbreviations used in this and related documents.

Table 1: Acronym Definitions

Acronym	Definition
YSM	Yourdon Structured Method
LTDC	Layered Transmission Display Controller
STM	STMicroelectronics
MEMS	Micro-Electro-Mechanical Systems
End of acronym definition table	

2.2 Glossary and Definitions

This section defines glossary terms used in this and related documents.

Table 2: Glossary Terms and Definitions

Glossary Term	Definition
STM32F429I	Micro-controller board has all component fit onto one board.
Customer	The professor that is view the grading all assignments.
End of glossary terms table	

2.3 Referenced Documents

This section lists the referenced documents for this document. The references are categorized into two categories:



External Documents not directly associated with this project.

Project Documents that are directly associated with this project.

2.3.1 External Documents

2.3.2 Project Specific Documents



CHAPTER 3

Required Work Overview

The scope of the work is to write software to an STM32F429I-Disc. The application would be a full stack development. This system shall require user input and display report. Therefore, that display would use a tools set just as touchgfx as an example. Touchgfx would be used to create a frontend generation of the system. The backend would run by a multi-threaded system.

3.1 Program Status

The project started because of the board that was purchased for another class. That class asked to create a system that requires input and outputs between an MCU. Specifically, an STM32 Board. It is easier to just use that project as a guideline to project plan for this class.

3.2 SDLC Situation

The software system would be composed with a multi-task system. This would be generated by autogen infrastructure. This combination between touch, GFX and STM 32 cube library.

3.3 Requirement Plans

The current plan is to write requirements once the deployment system is solidified. The requirements will be written in a different file. This file will be a csv that organizes what needs to be done for the system.

3.4 Documentation Plans

The following documents are listed here just to test reference generation. A “real” SDP would reference these as applicable for the project.

- `ref`KNEAD`OCD`ExProj, ref`KNEAD`OCD`ExProj [ref`KNEAD`OCD`ExProj]`
is the **OCD**, which outlines the project overall so, generally, it is created first.
- `ref`KNEAD`SDP`ExProj, ref`KNEAD`SDP`ExProj [ref`KNEAD`SDP`ExProj]`
is this document.
- `ref`KNEAD`SPS`ExProj, ref`KNEAD`SPS`ExProj [ref`KNEAD`SPS`ExProj]`



is the **SPS**, which should come from the customer or end user, but often is generated by the developer with customer approval.

- **ref`KNEAD`SSS`ExProj, ref`KNEAD`SSS`ExProj [ref`KNEAD`SSS`ExProj]**
is the **SSS** that is the developer's design specification to meet the **SPS** requirements.
- **ref`KNEAD`SUM`ExProj, ref`KNEAD`SUM`ExProj [ref`KNEAD`SUM`ExProj]**
is the **SUM** that acts somewhat like part of the **SSS** since it illustrates the **UI** design part of the **SSS**, but in a separate artifact that also can be used as a standalone users' manual.
- **ref`KNEAD`HRS`ExProj, ref`KNEAD`HRS`ExProj [ref`KNEAD`HRS`ExProj]**
is a **HRS**, which often is not used for smaller projects but can have multiple instances for large projects to more fully detail hardware design.
- **ref`KNEAD`SRS`ExProj, ref`KNEAD`SRS`ExProj [ref`KNEAD`SRS`ExProj]**
is a **SRS**, which often is not used for smaller projects but can have multiple instances for large projects to more fully detail software or firmware design.
- **ref`KNEAD`IRS`ExProj, ref`KNEAD`IRS`ExProj [ref`KNEAD`IRS`ExProj]**
is the **IRS**, which often is not use but may be needed, even if **HRS** or **SRS** artifacts are not, to fully document detailed interfaces such as Application Programming Interfaces (**APIs**) or other detailed mechanical or electrical interfaces.
- **ref`KNEAD`SSDD`ExProj, ref`KNEAD`SSDD`ExProj [ref`KNEAD`SSDD`ExProj]**
is the **SSDD** that provides a road map to the design and other design details needed to understand the hardware and software design.
- **ref`KNEAD`STP`ExProj, ref`KNEAD`STP`ExProj [ref`KNEAD`STP`ExProj]**
is the **STP** that highlights the planning for system testing.
- **ref`KNEAD`STS`ExProj, ref`KNEAD`STS`ExProj [ref`KNEAD`STS`ExProj]**
is the **STS**, which is sometimes called a test procedure. There could be multiple of these based on the overall project size.
- **ref`KNEAD`STR`ExProj, ref`KNEAD`STR`ExProj [ref`KNEAD`STR`ExProj]**
is an **STR** that documents the results of a given test. Multiple instances are expected



based on the test plan. And, there could be multiple versions of a given test plan to document repeated occurrences of a given test specification/procedure.

- **ref`KNEAD`SVD`ExProj, ref`KNEAD`SVD`ExProj [ref`KNEAD`SVD`ExProj]** is an **SVD** that documents a given release of a system. Multiple versions of these “release notes” are expected, with one **SVD** issued for each system release cycle.

3.5 Schedule and Resource Constraints

Mostly tools that I want to use for development or windows driven and I have a Mac so there’s a lot of overlay to figure out the best options to develop the software.

3.6 Other Constraints

Some other constraints would be the ability to develop a full system without proper material. Also being able to receive that material at the right time frame.



CHAPTER 4

System Development Plans

The current software plan is to use a combination of touch GFX and cube MX that all have a developer interaction to be able to produce the bugging code and release code.

4.1 Hardware Development Plans

The System will not need additional hardware for completion of work. If additional features are needed then this section shall be updated.

4.2 Firmware Development Plans

This System would require new updates to the firmware. Most firmware update is caused because of different packages that would update the middleware on the system. To keep the system up to date. Every release would have a tag to which firmware released.

4.3 Software Development Plans

The design shall be written as a uml format. Then would have a top down design.

4.4 Integration Plans

Currently, the plan is the update to the system through STLink. The STLink should be used to used to flash the system and debug the system.

4.5 Testing Plans

To keep a CICD pipeline structure, there shall be an automated process to build and test my code. As code get developed and processed every merge into a codebase.

4.6 Other Development Activities

There are non-other development activities needed.



CHAPTER 5

System Transition Plans

The system transition plan is to implement all the code on the host machine and then transfer it through pipelines that is driven from updates on our get library. These updates will then be automatically transferred into the STM 32 board that is connected to the host machine. Once all the changes are done, then the house machine and the board will have the most updated software on it.

5.1 Configuration Management Plans

The configuration management system is primarily using the standard sprint. The standard Sprint will be a one week basis instead of a two week basis, and all of the implementations will be written thoroughly per task creation.

5.2 Release Plans

The release plan is scheduled to send out new updated releases every single month, allowing the user to play with it and tell us how it works for them.

5.3 User Support Plans

They will be monthly releases for new software update updates for the system.

5.4 Other Transition Plans



CHAPTER 6

Management and Control Activities

The full codebase will be Version Source Controlled(VSC).

6.1 Technical Review Events

The development system does not have technical review events.

6.2 Skills and Resources Needed

Users shall have documentation and demonstrations on how the system would work. This would be displayed on the LCD screen.

6.3 Scheduled Development and Monitoring

This would be maintained by the professor's schedule.

6.4 Other Management and Control Activities



APPENDIX

Other Info

This section provides other information, as necessary, to document the system development plan.



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